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PLACEMENT DIRECTORS

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EDUCATION

Ph.D. in Economics, McGill University

expected May 2024

M.A. in Economics, Ewha Womans University

2018

B.A. in Economics, Ewha Womans University

2016

FIELDS OF SPECIALIZATION

Primary: Econometrics, Time-Series Analysis

Secondary: Financial Economics

DISSERTATION

Inference for Models with Mixed-Frequency Data

Committee: Prof. Sílvia Gonçalves (Chair), Prof. John Galbraith, Prof. Saraswata Chaudhuri

JOB MARKET PAPER

“Bootstrapping Factor-MIDAS Regression Models”.

Abstract: This paper studies the inference in factor-MIDAS regression models, a popular mixed-frequency estimation method. We derive the asymptotic distribution of the estimators under the assumption that the cross-sectional dimension is relatively small compared to the time series dimension. We find that there exists an asymptotic bias which depends not only on the cross-sectional dependence but also on the serial dependence of the idiosyncratic term in the factor model. To correct the bias, we propose a consistent plug-in bias estimator and a general residual-based bootstrap method. For the bootstrap method, we also provide a set of conditions under which any general residual-based bootstrap is valid. We verify these conditions for a particular bootstrap procedure, which can mimic the serial and cross-sectional dependence in the error term of the factor model. We use Monte Carlo simulation to confirm the bias in the finite samples and the performance of two bias correction methods. Our simulation results show that both methods capture the bias well. However, the bootstrap method outperforms the plug-in bias estimator for correcting the distortion in the coverage rate. We apply the factor-MIDAS regression model to a forecasting example where the quarterly U.S. real GDP growth rate is nowcasted with monthly factors. In this application, we find a significant bias, highlighting the importance of our bias correction method.

WORKING PAPERS

“Bootstrap Inference for Group Factor Models”, with Sílvia Gonçalves and Benoit Perron.

Abstract: Recently, Andreou et al. (2019) have proposed a test for common factors based on canonical correlations between factors estimated separately from each group. We propose and theoretically justify a simple bootstrap test that avoids the need to estimate the bias and variance of the canonical correlations explicitly. We verify these conditions for a wild bootstrap scheme similar to the one proposed in Gonçalves and Perron (2014). Simulation experiments show that this bootstrap approach leads to rejection rates closer to the nominal level in all of our designs compared to the asymptotic framework.

WORK IN PROGRESS

“ Bootstrapping Factor Models with Sparse VAR Idiosyncratic Errors”, with Sílvia Gonçalves and Benoit Perron.

“ Bootstrapping Prediction Intervals for Nonlinear Factor-Augmented Regression Models”.

TEACHING EXPERIENCE

McGill University - Lecturer

Mathematics for Economists (graduate)	<i>Summer 2023</i>
Economic Statistics (undergraduate)	<i>Spring 2023</i>
Mathematics for Economists (graduate)	<i>Summer 2022</i>
Econometrics 2 Honours (undergraduate)	<i>Winter 2022</i>

McGill University - Teaching Assistant

Introductory Econometrics I	<i>Fall 2023</i>
Macroeconomic Analysis & Application (undergraduate)	<i>Spring 2023</i>
Econometrics I Honours (undergraduate)	<i>Fall 2022</i>
Microeconomic Theory I (graduate)	<i>Fall 2021</i>
Macroeconomic Theory I (graduate)	<i>Fall 2021</i>
Applied Cross-Sectional Methods (graduate)	<i>Spring 2021</i>
Applied Time-Series & Forecast (graduate)	<i>Spring 2021</i>
Economic Statistics (undergraduate)	<i>Fall 2018 - Fall 2020</i>

Ewha Womans University - Teaching Assistant

Econometrics (graduate)	<i>Fall 2017</i>
Money and Banking (undergraduate)	<i>Fall 2017</i>
Financial Econometrics (undergraduate)	<i>Fall 2017</i>
Introduction to Financial Engineering (undergraduate)	<i>Fall 2017</i>
Financial Economics (undergraduate)	<i>Spring 2017</i>
Microeconometrics (graduate)	<i>Spring 2016</i>
Applied Econometrics (undergraduate)	<i>Spring 2016 - Spring 2017</i>
Statistics for Economics (undergraduate)	<i>Spring 2016 - Fall 2017</i>

CONFERENCE AND SEMINAR PRESENTATIONS

- 2023 Canadian Econometrics Study Group , October (Scheduled)
Econometric Society North American Summer Meeting, June
Society of Financial Econometrics, June
CIREQ Econometrics Conference, May
Virtual Workshop for Junior Researchers in Time Series, April
CIREQ, McGill University, April
- 2022 International Symposium on Econometric Theory and Applications, July
Canadian Economic Association, June
CIREQ, McGill University, April

GRANTS AND AWARDS

2018 - 2024 Graduate Excellence Award, McGill University

OTHER RESEARCH EXPERIENCE

Prof. Sílvia Gonçalves, McGill University	<i>Spring 2022</i>
Prof. Hosin Song, Ewha Womans University	<i>Spring - Winter 2017</i>
Prof. Vladimir Hlasny, Ewha Womans University	<i>Fall 2016 - Spring 2017</i>

COMPUTATIONAL SKILLS

Matlab, STATA, R, Python

LANGUAGES

English (Fluent), Korean (Native), French (Intermediate)

REFERENCES

Prof. Sílvia Gonçalves, McGill University	silvia.goncalves@mcgill.ca (514) 398-3392
Prof. Benoit Perron, Université de Montréal	benoit.perron@umontreal.ca (514) 434-5831
Prof. John Galbraith, McGill University	john.galbraith@mcgill.ca (514) 398-5167